

LLMServingSim — A40 Validation Report

Date: 2026-06-19 Model: Llama-3.1-8B / 70B HW: NVIDIA A40 x8 (48GB)

Backend: ASTRA-Sim analytical Workload: sharegpt 300 req, rate 10/s, fp16

Scope of work

- Profiled A40 (8B tp1/2, 70B tp4/tp8) incl. GPU power; extrapolated tp4/tp8.
- Validated simulator vs real vLLM on A40 for TP=1/2/4/8 (TTFT, TPOT, throughput, power).
- Found & fixed a profiler bug (record_function -> cuda_event).
- Studied extrapolation validity (tp8) and cross-hardware 70B (A40/A100/H100).

Key findings

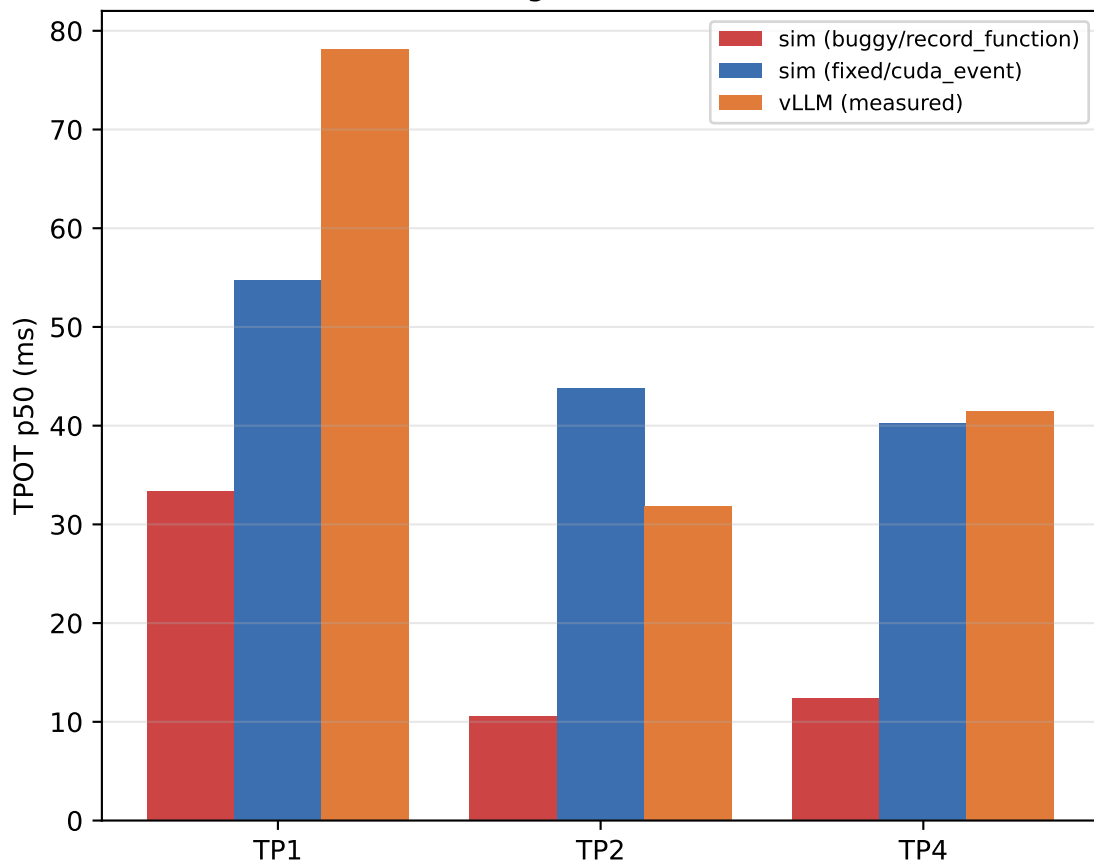
- GPU power matches vLLM within <1.5% at TP1/TP2 (sim active_power=300W ~ A40 TDP).
- Profiler bug (record_function) under-measured layer GEMMs AND attention (decode x6.65). Switching to cuda_event fixed it; TPOT match improved from -57..-70% to within +-3..38% (TP4 near-exact).
- Corrected sim is mildly conservative on throughput (-13..-26%).
- Extrapolation valid on compute side (end-to-end +-10..18%), but at high TP the limiting error is the sim's single-link collective model, not the profile:
real vLLM TP=8 collapses (690 tok/s, 18.8s TTFT) via cross-NUMA all-reduce;
the sim cannot reproduce it.
- 70B TP4 cross-hw: A100=2.5x, H100=3.2x A40 throughput; A100 best perf/J.

Measured interconnect (A40, P2P)

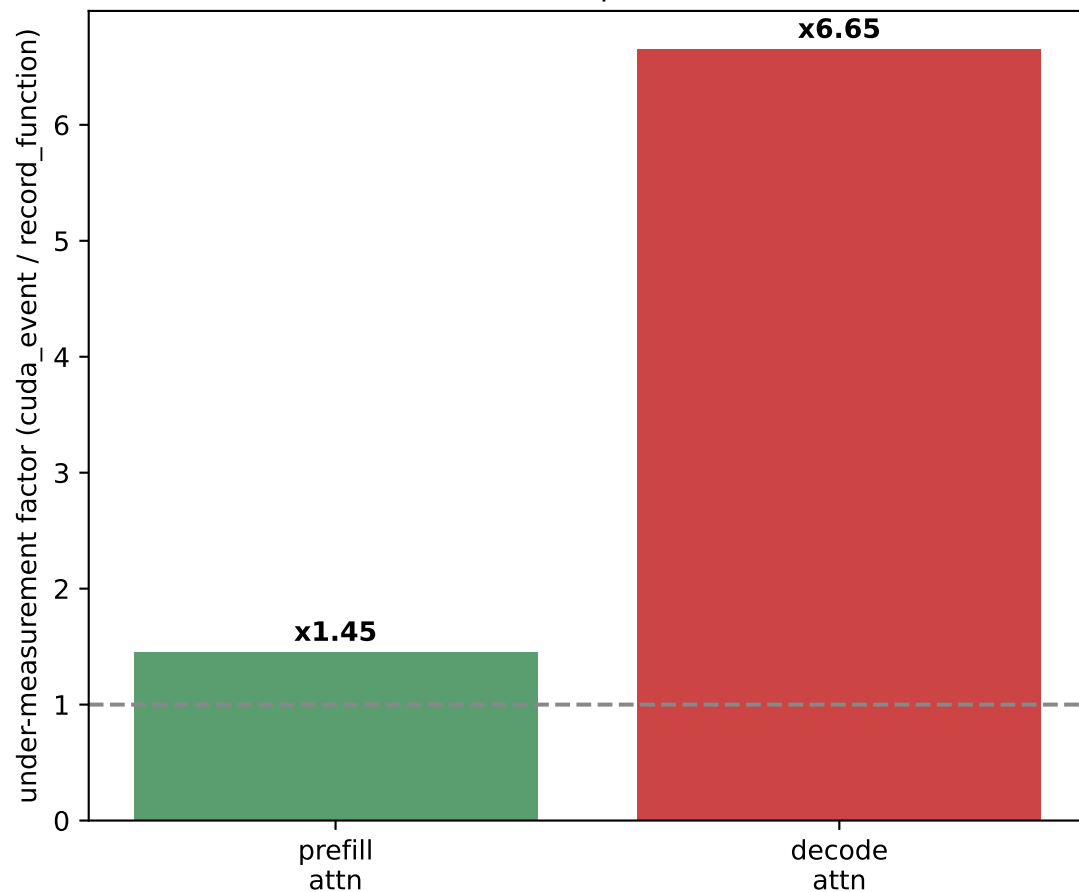
- NVLink pair (GPU0-1): 52.8 GB/s • PCIe cross-pair (0-2): 24.5 GB/s
- cross-NUMA (0-4, SYS): 21.0 GB/s (TP8 all-reduce bottleneck)

(1) Profiler bug: record_function -> cuda_event

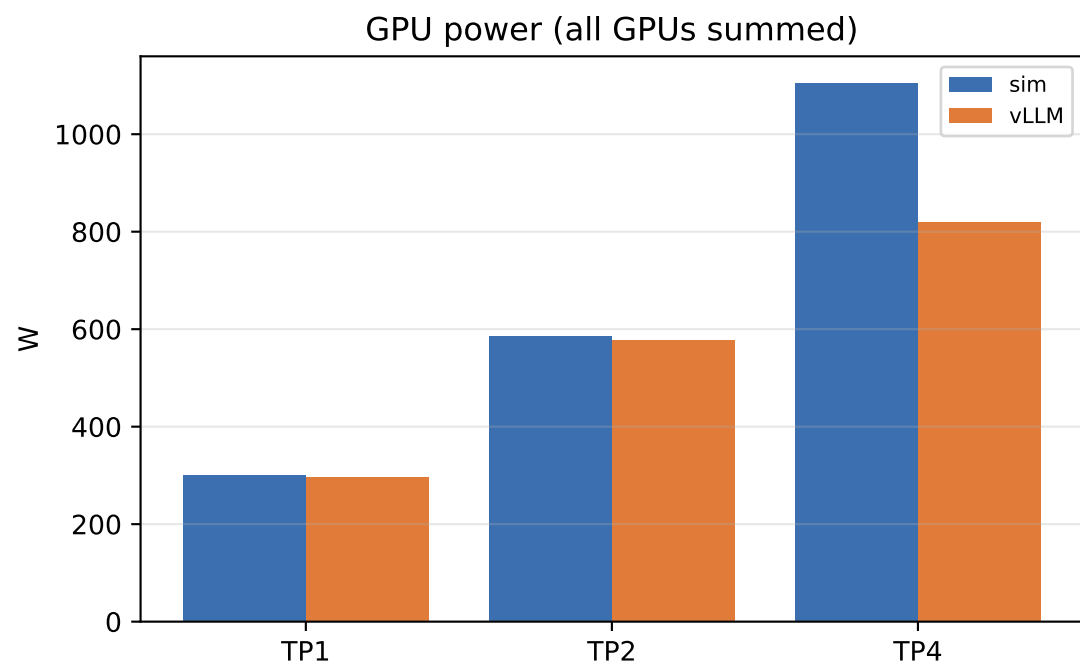
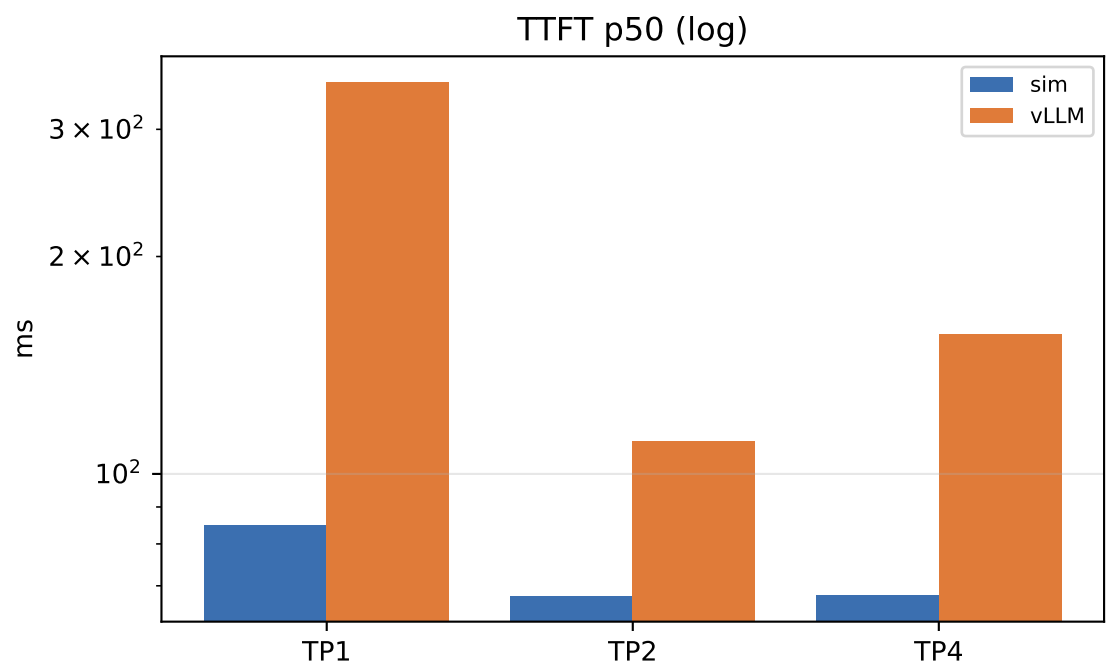
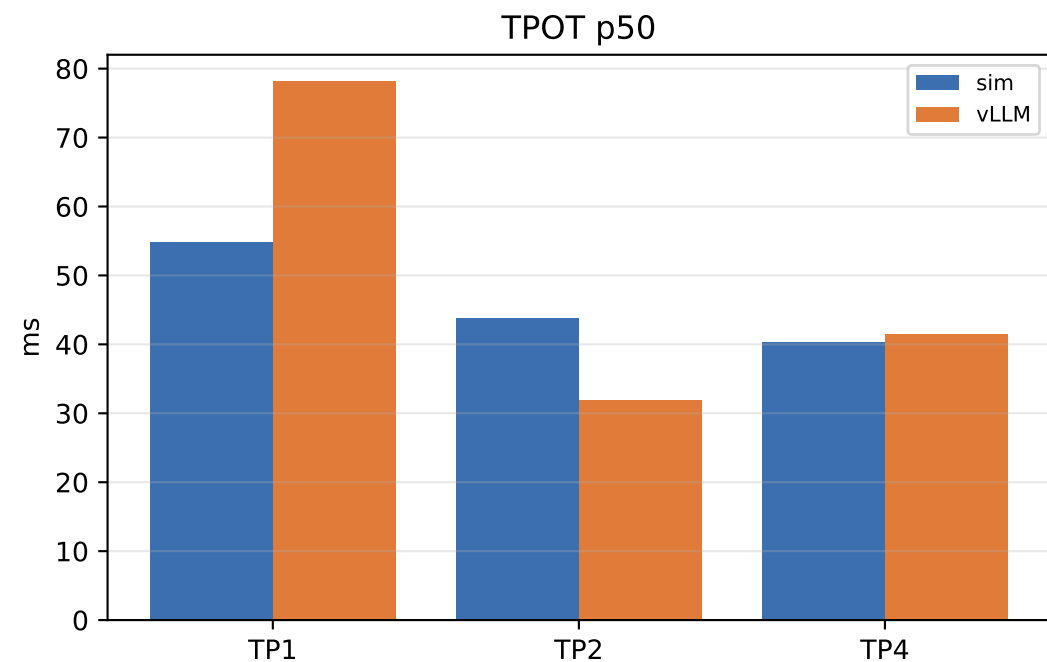
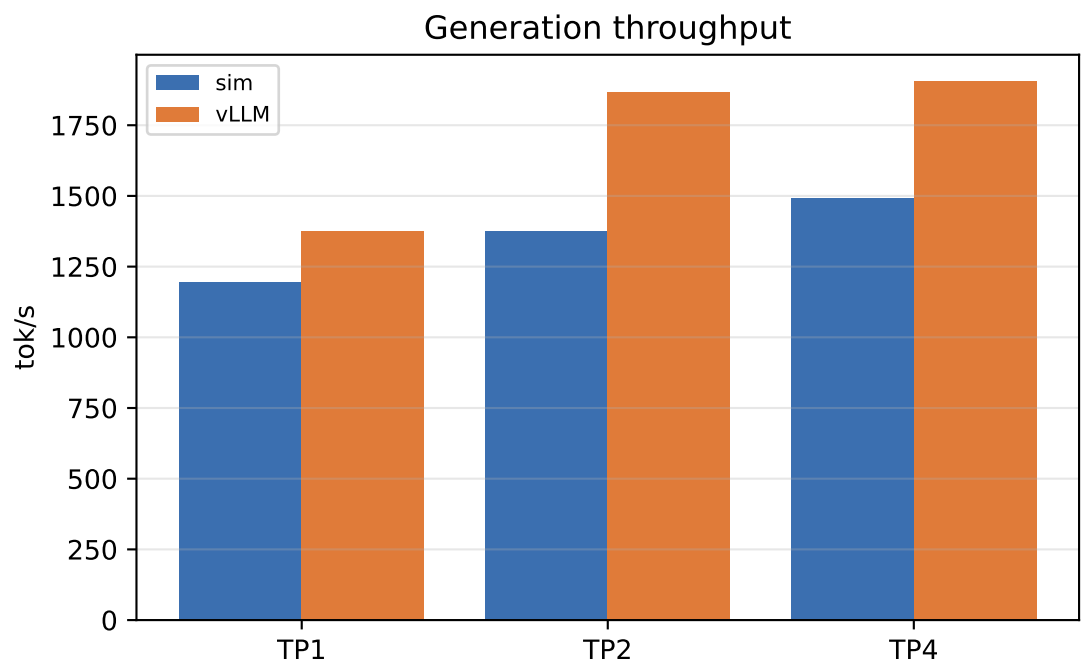
Profiler-bug fix: TPOT vs vLLM



record_function under-measured attention
(8B tp2)

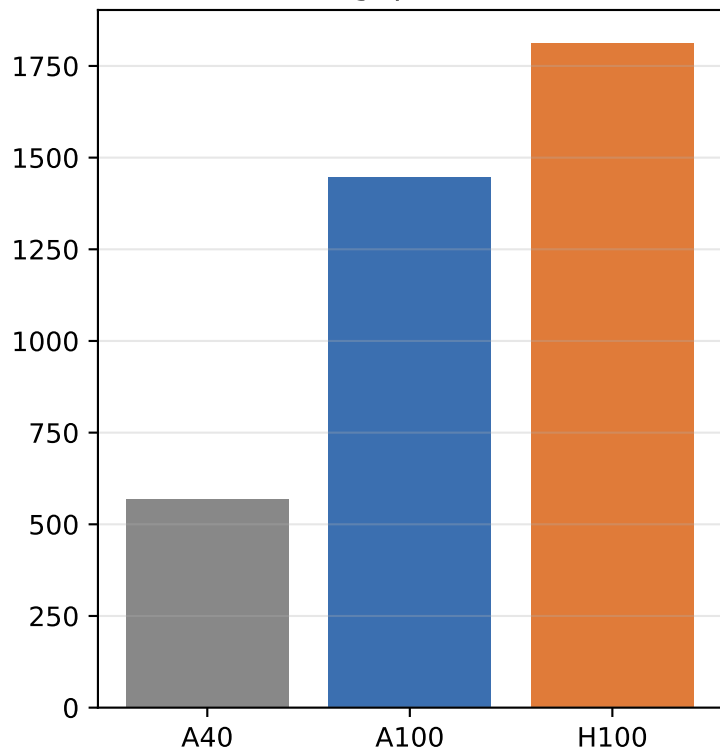


(2) Corrected sim vs vLLM — Llama-3.1-8B, A40, TP1/2/4

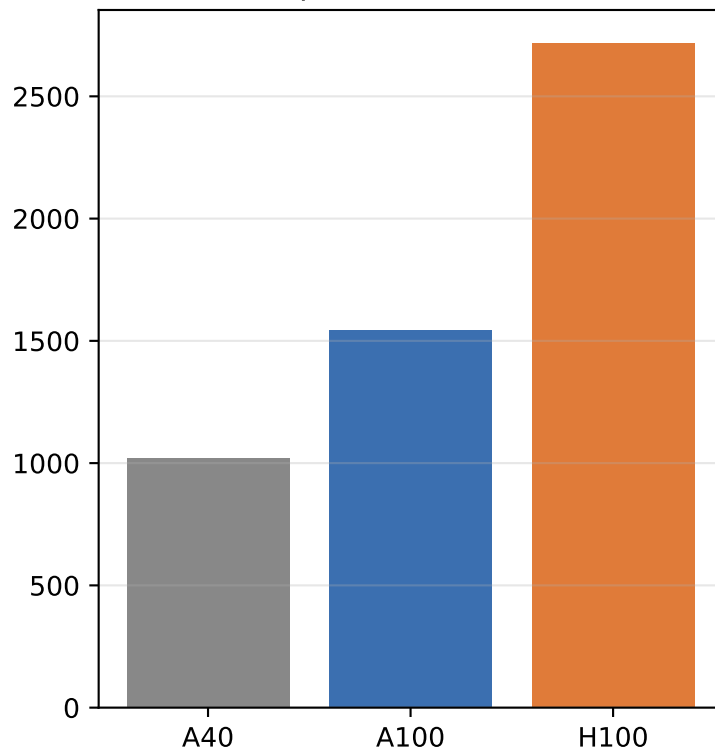


(3) Llama-3.1-70B TP=4 — cross-hardware sim (A40/A100/H100)

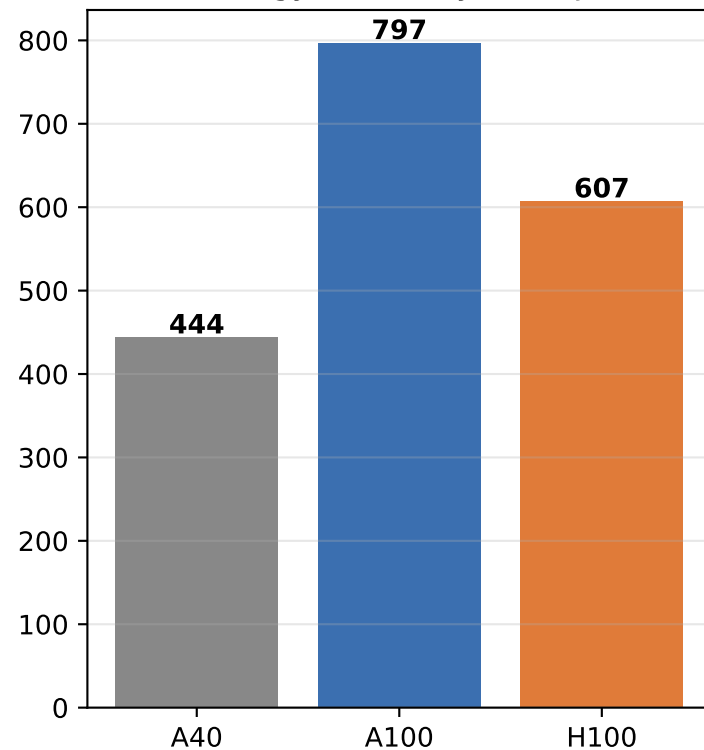
Throughput (tok/s)



GPU power (4 GPUs, W)



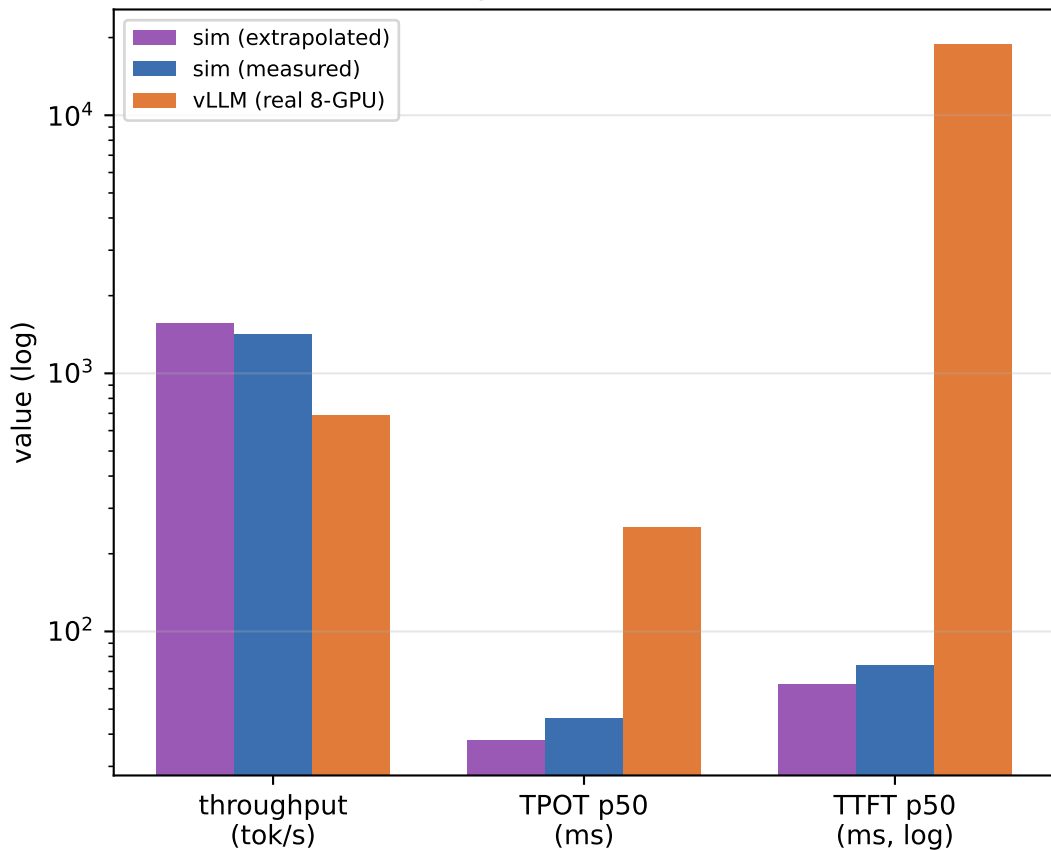
Energy efficiency (tok/kJ)



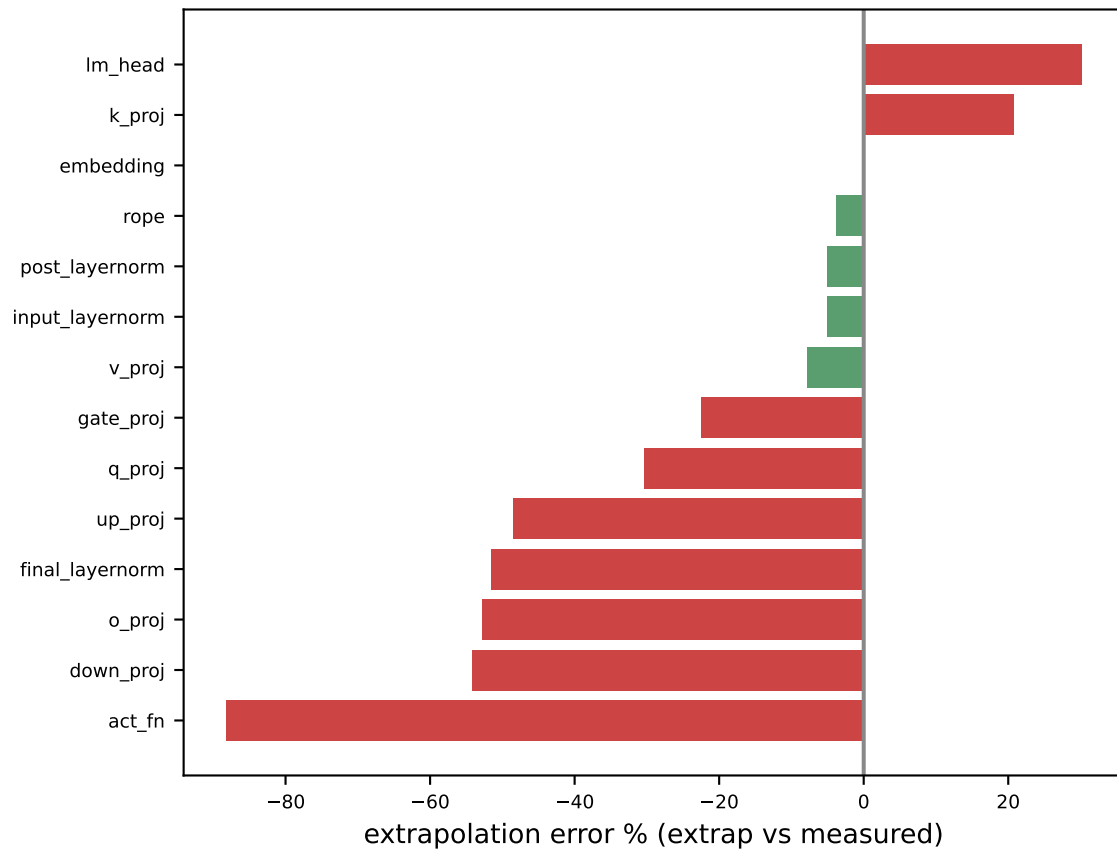
sim prediction; A40 link/power measured, A100/H100 from datasheet (estimates)

(4) TP=8 extrapolation validity (8B, 8x A40)

End-to-end: extrap vs measured vs vLLM (TP=8)



Per-layer tp8 extrap error
MAE=30%



vLLM TP=8 collapses (cross-NUMA all-reduce); sim can't reproduce -> high-TP limit is the sim collective model, not extrapolation

(5) A40 Llama-3.1-8B — sim vs vLLM (ms / tok/s / W) [TP8 sim=measured profile]

Config	sim TTFT	vLLM TTFT	sim TPOT	vLLM TPOT	sim tput	vLLM tput	sim pow	vLLM pow
8B TP1	85	349	54.8	78.1	1195	1374	300	297
8B TP2	68	111	43.8	31.8	1374	1864	586	578
8B TP4	68	156	40.3	41.5	1491	1904	1105	820
8B TP8	74	18832	46.0	252.5	1422	690	-	-